

Improving Agile Team Learning by Improving Team Reflections

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Abstract

Many Agile proponents encourage reflection as part of the feedback/learning cycle. When we began using Agile practices, including reflection, I discovered that our organization's norms and beliefs tended to discount the value of regular reflection meetings. Many engineers wanted to avoid anything that might be seen as "touchy-feely." Our initial attempts at holding regular reflection meetings for agile teams had mixed results. Some reflections turned into gripe sessions where a lot of venting occurred but nothing changed. Other reflections were short and a little too sweet – team members said little and no learning occurred. This paper describes some of our efforts helping teams improve their reflection meetings and relates some of the positive results in my organization as a result of regular reflection. It also discusses techniques for overcoming initial reluctance to take time out to reflect, keeping reflection meetings interesting, and taking reflection to a deeper level.

1. Introduction.

Regular reflection is an Agile practice that can improve teamwork, productivity, and group learning. Reflecting on our experiences individually and in a group allows us to learn and improve. Many authors of Agile Software Development books list reflection as an important part of the feedback and learning cycle that Agile practices rely on. In our organization, the practice of reflection has been one of the most challenging Agile practices to implement. Even though it has been challenging, our experiences with reflection have helped us to improve our team's productivity, quality, and collaboration.

I have worked on an Agile project for a little over 2 years and I have learned a lot about what works and what doesn't work in doing reflection. I hope that by relating our experiences, including our mistakes and our successes, others will persist in encouraging reflection with their teams.

2. The Value of Reflection.

Regular reflection is important for many reasons. Not only do our teams reflect on what they are doing, but they also reflect on how they are working together as a team. Reflecting together helps team members get to know one another better, surfaces conflict, and develops a higher degree of trust.

On our team, reflection meetings help us learn together and to apply our learning to improving our practices. Our project was one of the first to adopt Agile practices in our organization. Because of the size of the project (>20 developers) and the unique needs of our product development environment we found that using a try/reflect/adapt cycle helped us learn which practices worked, which practices didn't work, and how to adapt practices to make them work better in our environment. In addition to helping us improve our practices, I believe that reflection has helped teams resolve conflicts and improve team learning. I have also seen significant innovation and process improvements as a direct result of reflections. These innovations and improvements have created significant productivity gains for our team.

3. Getting started can be difficult.

I still encounter strong resistance from some developers and managers when I encourage teams to reflect regularly. Some of their questions include, "What is the outcome of a reflection meeting?" and "Isn't this just another way to decrease velocity?" I have also heard comments from developers like, "It's just a waste of my time. I have real work to do," and "We tried reflection but it didn't work."

There were many reasons for this resistance and these beliefs. Like many software development organizations, Medtronic's culture is action oriented. There is a belief that in order to be effective, meetings should result in an action plan, a decision, or a list of problems requiring investigation. Even though we are working more collaboratively than ever, many developers still believe that attending meetings is not "real work." As project pressures increase, it's easy to put off reflection. We

don't have data yet that shows the cost effectiveness of reflection. And because the effects of reflection can be intangible and difficult to quantify, some managers do not understand or support it.

I believe that it is important for someone in the organization to recognize the value of reflection and to advocate for it. In our organization, I have become an advocate of reflection. In the beginning, as project architect, I had the opportunity to sit in on daily meetings, planning meetings, and reflection meetings for all of the teams on our project. I could see what was and what wasn't working and make suggestions for improving the quality of reflection meetings. I have also facilitated reflection meetings for teams on our project and for other projects in our organization. I also encourage managers and team coaches to be open to trying reflection, even if they aren't convinced of its value.

When managers or developers question the value of reflection, I cite improvements that I have seen as a result of reflection meetings. I also use a quote a consultant shared with me, "There is no learning without conversation."

Not all Agile teams in our organization reflect regularly. Our project is the only one that I am aware of that holds regular reflection meetings. Others have tried holding reflection meetings and encountered many of the same roadblocks that we encountered early on. Because there wasn't someone advocating for the practice of reflection and working to try new techniques to improve it, the teams dropped the practice.

4. Roadblocks to Effective Reflection

Our initial attempts at reflection had mixed results. We spent a lot of time complaining but little time in constructive conversation. I learned that while some venting may be needed, too much time spent venting has no positive result. Other reflections were short and team members said little.

Our first reflections occurred in conjunction with sprint planning meetings. We found that with the combined reflection and planning meetings, teams spent either too much time reflecting and didn't have time to complete their planning, or they spent all their time planning and didn't have time to reflect. Rather than extend the meeting time, we decided to separate planning and reflection activities. While there is a benefit to applying learning from reflection into the next sprint's planning, this can be accomplished with a reflection meeting the day before the planning meeting. With a two-week sprint, we found that most of our teams prefer to reflect during the week in which planning meetings do not occur.

Here is a list of some of the "mistakes" that we made when first doing reflection:

- Inviting managers and other "chickens" to observe or participate in a new team's reflection meetings. We learned that until a team has time to build trust and to gain practice reflecting together, including managers or outsiders inhibited free exchange of ideas.
- Trying to do a reflection with the entire development team (over 20 developers and 8-10 chickens in the room) when the team was actually divided into 4 smaller sub-teams. Everyone felt this was a waste of time and had the effect of creating negative attitudes about reflection which took some time to overcome.
- Repeating the "What went well that we want to keep doing? What needs improvement?" format meeting after meeting. After a couple of repetitions, the questions ran out of juice and it was all too easy to believe that there was nothing new to improve.
- Expecting a list of action items to result from each reflection meeting. Sometimes teams uncover problems and it is appropriate to develop an action list. Often, the learning is more subtle. Team members may come away with a better understanding of one another or something that happened in the meeting may trigger a great idea later.
- Allowing team members to choose not to attend reflection meetings. I worked with a team member who hated meetings of any kind. He complained so bitterly about reflection meetings that I suggested that he could choose whether to attend meetings. After he chose not to show up at two reflection meetings in a row teammates became very unhappy. They found that they had unfinished business that could not be completed with one team member absent. The team requested that he attend their reflection meetings. Resolving conflicts and talking about how the team is working together require everyone's input.
- Giving up on reflection after one or two tries. Another project at Medtronic began holding regular reflection meetings for its team. They used the "What is going well? What can be improved" format. After a couple of meetings where no new ideas were uncovered team members decided that reflection was a waste of time and stopped the practice. Learning to reflect is a process that takes persistence, practice, feedback, and adaptation.
- Holding team reflection meetings without a leader. Recently the coach of a team was absent from a reflection meeting. No one was designated to lead

the reflection in his absence. Instead of reflecting, the team began planning the next phase of work they were doing. A customer who had attended the meeting told me that she had not realized the importance of having a facilitator for a reflection meeting until she attended one without a facilitator.

5. A Technique for Improving Reflection

As we encountered many of the roadblocks to effective reflection, I wanted to find a way to improve these meetings. I tried a technique I had experienced at a conference. The technique, called Conversation Café, combines simple ground rules with the use of a talking token to help guide conversations. It encourages all to participate and listen, creates a safe structure for conversation, and helps teams develop trust.

The simple ground rules of the Conversation Café method provide reminders of the qualities of good conversation—acceptance, listening with respect, curiosity, diversity, sincerity, and brevity. The use of a talking token allows everyone's voice to be heard. And the experience of being heard and moving conversations to a deeper level is helping to convince developers that there is value in conversations, even when they do not result in immediate action items or decisions.

6. The Conversation Café Technique

The Conversation Café technique consists of six agreements and a meeting process. According to the Conversation Café website, "Conversation Cafés are places where collective learning is happening. Conversation Cafés are not instead of action. They are before action—a place to gather your thoughts, find your natural allies, discover your blind spots, open your heart to the heart of 'the other.' All movements begin in conversation."

7. Conversation Café Agreements

- **Acceptance:** Suspend judgment as best you can.
- **Listening:** With respect
- **Curiosity:** Seek to understand rather than persuade.
- **Diversity:** Invite and honor all points of view.
- **Sincerity:** Speak what has personal heart and meaning.
- **Brevity:** Go for honesty and depth but don't go on and on.

8. Conversation Café Process

- **Assemble** up to 8 people plus host, hearty topic, "talking object," and time (60-90 min.). Host explains process and agreements.
- **Round 1:** Pass around the talking object; each person speaks briefly to the topic, no feedback or response.
- **Round 2:** Again with the talking object; each person deepens their own comments or speaks what has meaning now.
- **Dialogue:** Open, spirited conversation. Use talking object if there is domination, contention, or lack of focus.
- **Final Round:** With talking object, each person says briefly what was meaningful to them.

For more information about Conversation Café, visit www.conversationcafe.org.

9. My First Experience Using Conversation Café

The first time I used the Conversation Café method with one of our teams, I had a sensitive topic I wanted to discuss. While participating in several design meetings with a team, I had noticed a number of errors in the team's code. While the team's review process enabled us to find the errors, I wanted to explore my concerns about how we could prevent such errors from occurring and explore what kinds of learning needed to take place for the team. With the team coach's support, I decided to use the Conversation Café method with the team during their next reflection meeting.

At the meeting I introduced the ground rules, the process we would follow, and the talking object to the team. I set an example of speaking with honesty and heart. Talking piece in hand, I described what I had seen and the concerns that I had. I wondered out loud whether anyone else had concerns and passed the talking piece to the next person. One by one, every team member but one expressed similar concerns. I was stunned. After two rounds of using the talking piece, we opened up the discussion and the team began to explore possible actions that they could take to alleviate the problem. By the end of the reflection meeting, they had chosen three actions to take and the team went on to improve their processes and learning together significantly.

10. Success of the Technique

I knew that I had found a technique that worked when I used the technique with a second team. Both teams asked if we could use the talking piece reflection method again. I believe that using this method in many of our

early reflections improved the quality of reflection meetings tremendously and taught our teams some of the conversational practices that open minds and hearts to new ideas. We don't start each reflection meeting with a specific topic, but almost always, we find a topic that resonates with the team.

I have found that using this method helps us to delve deeper and helps topics to emerge that might not otherwise arise. It also introduces learning that impacts group behavior and the process and practices that the teams follow.

I was recently in a reflection meeting where we were not using a talking piece. The conversation began to get very lively and fast. When one of the participants got interrupted, he exclaimed, "Where's our talking pen?" (At the last reflection meeting we had used a pen as a talking piece.) The conversation slowed down. The speaker was able to continue uninterrupted and the meeting proceeded.

Recently a coach told me that facilitating reflection meetings was one of the hardest parts of the coach role. Using the Conversation Café process helped her to become more comfortable leading reflection meetings.

I have to admit that I was surprised by how well the teams accepted this practice. It took great courage for me to introduce this technique the first time. Later when I described it to a manager in our area, he was incredulous that the method worked. His comment was memorable, "You did what? With engineers? And they liked it?"

11. Conversation Café Cautions

While I have found the Conversation Café technique helpful, it is not a panacea. Teams can still have unproductive reflection meetings using this technique. The first person to speak often determines the direction the meeting will take, especially if that person is the perceived leader of the meeting.

Often, the other individuals in the group will learn by observing the first person to speak, what the norm and expectation of the group is for honesty and depth. The honesty and depth of the reflections of the team are directly related to the honesty and depth the leader is willing to express.

If I am leading a team reflection and I succumb to "Minnesota nice," then the team members will almost always stay in the safety zone of "Minnesota nice" as well. I have led reflections on topics that should have had some juice, but because I had difficulty going for honesty and depth myself, no one else did either. I can tell when I am in a reflection like that because the meeting feels dead and there is no "energy" in the group.

Another mistake I have to school myself to avoid, is jumping in when the group has an initial slow start.

Sometimes a team will start out by saying that they have nothing to discuss. By being patient, sitting through some possible silence and letting the meeting continue, I often find that either the group happens upon a topic accidentally that yields significant learning, or someone feels safe enough to bring up a topic of concern. Expected behavior may also be at work. A long silence with the expectation that someone will speak builds tension that has to be released.

12. Evolve Reflection Meeting Techniques to Keep them Fresh

As time goes by I use the Conversation Café technique less and less overtly. I have noticed that once teams become adept with this process they don't need to use a talking piece and follow the steps so explicitly. When I notice that reflections have become dull and routine, I will bring the talking piece to a reflection meeting and use the method again. I suspect that if we used the Conversation Café technique for every reflection meeting, that it too would lose its effectiveness.

When you use Conversation Café, don't forget to go over the ground rules and process, even if you think that the team knows them. The ground rules remind us of good practices to use any time we are in a conversation and it helps to remind people of the process in case they have forgotten.

I recently observed reflections of four different sub-teams on our project. Of the four teams I observed, two were using a talking token at their meetings. Each team's reflection was different and each team's reflection was effective for that team. One team with a new team member spent time introducing themselves to the new team member and learning about their new teammate. Another team discussed problems they were experiencing with their customer and developed a plan to resolve the problem. The team that I coach discussed the results of a recent survey on the team's Agile practices. During the discussion, they explored the reasons that the team's ratings changed and asked for more information on some of the practices that they didn't fully understand.

13. Other Factors that Influence the Quality of Reflection Meetings

There are other factors to consider when planning effective reflection meetings. When a team is first reflecting together, they do better without outside observers. Once they are accustomed to reflecting together, this rule can be relaxed, but initially when the group is working on developing trust, outsiders hinder the process. It also helps to set up ground rules for the

process. Many of our teams have agreed that what happens in the reflection stays within the team unless they agree to talk about it outside the team.

The size of group is also very important. Because our project team is large (over 20 developers), we have divided the team into smaller sub-teams that work together, meet together, and reflect together. We have found that teams of 4-6 developers seem to be the optimal size in our environment. If your team is larger than 8 people, consider dividing into smaller groups for a portion of the reflection meeting and then coming back together and debriefing the small group conversations with the entire group.

The shape of the meeting space also impacts the quality of reflections. I think the ideal shape for reflection is a circle of chairs with no table present. Second best is a team sitting around a table. If the table is too large or u-shaped, the quality of the reflection is usually impacted negatively.

14. Stories about Reflection Impacts

Since the introduction of Conversation Café, our teams have created their own reflection meeting innovations and explored many areas. After being asked to fill out feedback forms on each teammate by the project's leadership, one team decided that they would like to give feedback to one another face-to-face on how they perceived one another's performance. While it wasn't easy, they each shared one contribution to the team each teammate made that they most appreciated and one way that each teammate could improve how they worked with the team. One developer told me that the feedback he received was not what he had expected, yet it was "dead on" and helped him to see how his choices impacted the team. He also told me that this reflection made a palpable difference in how the team worked together. After giving one another feedback they were able to say things to one another that they would not have said prior to giving one another feedback. This team had been working together and reflecting together for almost a year before they held this meeting. Trying this technique with a newly formed team might not yield the same positive result. Imposing this topic on a team would also not yield a good result. Because the idea came from within the team and team members had developed sufficient trust in one another it worked.

The coach of that team later told me that he had found that the best reflections for his team seemed to flow from a topic that was currently "up" for someone on the team, and the least effective reflections happened when he introduced a topic that came from outside the team.

In one of my team's reflection meetings, I started a meeting with a simple, "Does anyone have anything they

want to talk about today?" One of the team members quickly jumped in to say that she had something she wanted to discuss. She went on to describe how difficult it was pairing with one of the senior members on the team. She related that as a relatively new developer, she was in learning mode and needed time to ask questions, slow down and think about the code, and try things on her own. She found it very frustrating when pairing with a teammate who was very skillful and fast because she did not feel that she could slow the process down enough to learn at a pace that was comfortable for her. She wanted to discuss what could be done to improve her pairing experience. As a result of her simple act of courage in expressing frustration with the pairing experience, the team went on to have a very open and honest conversation about pairing. While there were no easy answers, having the discussion made everyone more aware of the challenges and a little more considerate of one another as they worked together.

Early in our project, hearing complaints in a reflection meeting about a technical problem that everyone was encountering led one developer to think about the problem in more depth. She decided that solving this problem would be of great benefit to the project. With the support of the project manager, she convened a group of volunteers who were interested in working on the problem. Over a period of weeks the group met to explore various options. They developed and implemented a solution that has saved our project countless hours of effort. It is difficult to quantify how changes like this impact a project, but I believe that we have seen more innovation and new ideas on this project than on any project I have worked on. I attribute some of this innovation to increasing opportunities for developers to reflect together.

I am often surprised by the topics that come up in reflection meetings. At my team's most recent reflection, I wondered if anything would pop as an issue. The team had just finished a good sprint. Team members were getting along well and everything seemed to be going smoothly. As I started the meeting with, "What's up for everyone? Does anyone have anything they want to discuss today?" words and body language indicated that no one had much to say. As I let the silence sit for a couple of minutes, someone made a comment about a technical issue with a task he and a teammate were working on. This led to a short technical discussion and led to more questions. Before I knew it, another team member noticed that it appeared from the discussion that a pair was not following practices that the team had agreed on. Since the team had agreed that ignoring this practice had caused problems in a previous reflection meeting, they valued it enough to explore further. As they discussed the issue, with one another, they worked to

find a solution that met the needs of the pair and at the same time followed the practices that they had agreed on as a team. To me, this was a perfect example of the team learning that can occur when teams take time to reflect.

15. Conclusion

We learn from the past through cycles of action and reflection that lead to new actions. Without setting aside time for reflection and learning how to reflect well together, teams are missing an important learning opportunity. The practice of reflection has helped our Agile teams become more effective and productive. I hope our experiences with reflection meetings inspire you to give the practice of reflection a try.